Engineering Applications of Artificial Intelligence 161 (2025) 112203

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Spectrum prior-based and visibility fusion method for underwater image enhancement

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ARTICLE INFO

Keywords:
Underwater image enhancement
Hybrid fusion algorithm
Color correction
Visibility enhancement
Dehazing module

ABSTRACT

When light propagates in water, it undergoes scattering and absorption phenomena, which typically result in haze, high blur, low contrast and color distortion, making it extremely challenging to obtain high-quality images. To address these issues, many existing methods target image enhancement by correcting specific aspects such as color shift or contrast. However, challenges like poor visibility and low-light conditions are often overlooked. In this paper, we proposed a spectrum prior-based and visibility fusion method (SPV) to enhance underwater images in terms of color, contrast, and visibility. Unlike existing methods, SPV complements the advantages of both physical and non-physical models, comprehensively addressing the problems of reduced visual visibility, color distortion, and low contrast caused by low-light environments, thereby significantly improving the overall image quality. We proposed a dehazing module based on spectral information priors, which reliably restores image quality under complex water conditions. Additionally, we introduced a color correction module based on human color perception and employed morphological operations, effectively solving the issues of color shift and unclear contours in underwater images. Furthermore, we proposed a visibility enhancement module based on the fuzzy c-means clustering method to improve image contrast and visibility, particularly under low-light conditions. Finally, through a detail enhancement fusion module, we simultaneously addressed problems related to color shift, low contrast, and low visibility. SPV showed excellent performance in application tests including feature point matching, geometric rotation estimation, and edge detection. Comparative experiments on four real underwater datasets against 14 advanced enhancement methods demonstrated promising results.

1. Introduction

During underwater filming, images often encounter problems such as color shifts, reduced visibility, and decreased contrast. This is because light undergoes selective attenuation as it propagates through water, with different wavelengths being absorbed to varying degrees, resulting in color shifts. Additionally, tiny particles in the water scatter light, significantly reducing visibility. Moreover, the combined effects of scattering and absorption of light during its propagation in water lead to decreased image contrast. These factors collectively result in a significant degradation in the quality of underwater images, causing serious negative impacts across various engineering fields. In marine science, degraded images can make species identification difficult (Kaur and Vijay, 2022), distort environmental monitoring (Kong et al., 2022), and compromise geological research accuracy (Chen et al., 2021). In detection and monitoring, inaccurate structural inspections, ineffective pollution monitoring, and incomplete ecological assessments may occur. Therefore, employing effective under image enhancement methods can improve the color accuracy, enhance visibility, and increase contrast of underwater images, yielding significant practical benefits across multiple fields such as scientific research and engineering applications.

To address the problem of underwater image degradation, physical model-based methods (Mishra et al., 2024; Hou et al., 2023; Zhou et al., 2022a; Liang et al., 2021a; Wang et al., 2020) have achieved remarkable success. For example, Li et al. (2022a) proposed a method that derives the three-channel transmission relationship based on global background light and inherent optical properties and established a local transmission optimization model for underwater image restoration. These methods

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https://doi.org/10.1016/j.engappai.2025.112203
Received 27 June 2024; Received in revised form 5 July 2025; Accepted 1 September 2025
Available online 11 September 2025

model the scattering and absorption processes of light propagation underwater. Through the estimation of transmission maps and ambient light, the atmospheric light model is inverted to restore the original image. This approach effectively corrects color shifts, enhances contrast and clarity, and preserves more image details. However, physical model-based methods often require prior information or specific environmental parameters to accurately estimate transmission rates and ambient light, which can be challenging to obtain in practical applications. Additionally, the robustness and adaptability of the algorithms need to be further improved to ensure consistent dehazing effects under different types of water and shooting conditions.

Non-physical model-based methods (Lin et al., 2023; Qiang et al., 2024; Xu et al., 2023; Zhang et al., 2023a) have shown superior performance in enhancing underwater images. For example, Zhang et al. (2022a) proposed an underwater image enhancement method based on Retinex color correction and detail-preserving fusion techniques. By removing color cast through color correction and enhancing local contrast, details, and global contrast versions, the method improves image quality. These methods employ mathematical and statistical techniques to enhance image quality without relying on physical models of light propagation. By directly adjusting pixel values, they effectively improve contrast, visibility, color fidelity, and detail preservation. However, these methods are usually based on fixed rules and algorithms, lacking adaptive capability. Different images may require different enhancement parameters, and finding the optimal parameter combination often requires multiple trials.

Due to the complexity of underwater image degradation, which encompasses factors such as light absorption and scattering, interference from suspended particles (Khan et al., 2016) in the water, variations in lighting conditions, and water disturbances (Chen et al., 2019), any single method is difficult to address all these issues. Physical models face limitations in highly turbid waters, multi-light source interference, and dynamic scenes, and often perform unstably under low-light conditions, leading to issues such as color distortion, over-enhancement, and poor adaptability. Underwater image enhancement algorithms based on non-physical models commonly suffer from color distortion, excessive enhancement, poor adaptability, and a strong dependence on parameter tuning (Wang et al., 2024). To achieve more effective underwater image enhancement, it is essential to integrate physical and non-physical models, leveraging their complementary strengths. The proposed method combines the precise modeling of underwater light attenuation with the flexibility and adaptability of non-physical models for low-light image processing which effectively mitigates common degradations in low-light environments such as reduced visibility, color distortion, and low contrast, significantly enhancing overall image quality and robustness.

In this work, we proposed a hybrid method combining physical and non-physical models to leverage the advantages of different approaches while addressing the issues of color cast, low visibility, and low contrast in underwater images. Firstly, to achieve the effect of preliminary dehazing and address the challenge of image enhancement under complex water conditions, we adopt a physical model method based on spectral priors. By incorporating spectral priors, we can accurately estimate the transmission map without considering the differences in the transmission rates of light at different wavelengths in water (Chiang and Chen, 2011a). This results in more significant and stable image enhancement effects under complex water conditions. To correct color cast, we introduced a color correction module based on the CIELAB (Zhang and Wandell, 1996) color space, which enables accurate color adjustment by simulating biological visual mechanisms. Additionally, we employed morphological operations (Raid et al., 2014) to enhance object contour information in the images, making the object outlines clearer and improving the overall image quality and recognizability. Meanwhile, to tackle the issue of low visibility in underwater images, especially in low-light environments, we proposed a visibility enhancement method based on a fuzzy c-means clustering approach. When dealing with nearly invisible images, we enhance each pixel to varying degrees based on its fuzzy membership to different clusters, while preserving the original color and brightness information of the image. Finally, we proposed a fusion method that comprehensively enhances color fidelity, contrast, and visibility by combining images processed through color correction and visibility enhancement. The fusion module adaptively determines weights based on image gradient and luminance, effectively enhancing structural details and brightness in poorly illuminated regions.

The proposed method effectively enhances the visibility and contrast of underwater images, especially under low-light conditions. Surprisingly, SPV outperformed the advanced and typical approaches based on deep learning, physical models, and non-physical models. Fig. 1 presents a visual comparison between the SPV algorithm and other underwater image enhancement methods. The SPV algorithm maintains high contrast, high visibility, and rich color saturation, whereas other methods exhibit low contrast, poor visibility, color distortion, and inadequate detail performance. These results indicate significant advantages of proposed method. The main contributions of this work are as follows:

- We propose a fusion method that leverages the pixel intensity and global gradient of two enhanced images to achieve complementary advantages of physical and non-physical models. This module addresses the limitations of single methods, including insufficient detail restoration and instability under low-light conditions, as well as challenges related to over-enhancement and parameter sensitivity.
- To address blurriness and low contrast in underwater images, we propose a physical model-based enhancement method leveraging spectral information priors. This approach accurately estimates underwater transmission rates and ambient light, even in complex water conditions, thereby restoring image clarity and contrast while preserving natural and high-quality visual effects.
- Addressing the problem of low visibility in underwater images caused by absorption and scattering, especially under extremely low-light conditions, we propose a visibility enhancement module based on fuzzy c-means clustering. This module estimates and enhances the fuzzy membership of pixels to different clusters in the image while preserving color and brightness information, significantly improving visibility and naturalness in complex underwater environments.
- We conduct experiments on four benchmark datasets to evaluate the SPV algorithm using five no-reference evaluation metrics. Additionally, we test the algorithm's potential underwater applications, including edge detection, SIFT feature point matching, and geometric transformation estimation. These test results further validate the effectiveness of the SPV algorithm in practical applications underwater.

Because of absorption and scattering during the propagation of light underwater, a series of problems such as color shifts, reduced contrast, and decreased visibility arise. These issues severely impact the quality of underwater images, making it challenging to accurately depict the real underwater environment. In this paper, we propose a method to address the problems. Section 1 introduces the research background, research questions, current research status, and research significance, while also pointing out the pain points and challenges encountered by existing methods. Section 2 provides a detailed overview of the latest methods for underwater image enhancement. Section 3 elaborates on the details and principles of the modules proposed in our method, the spectral prior dehazing module, including the color correction module, the visibility enhancement module, and the detail enhancement fusion module. Section 4 describes the experimental setup, comparative experiments, and ablation experiments conducted on four datasets, as well as application tests. Section 5 summarizes the entire paper, discussing the advantages, limitations, and future research directions of the proposed method based on experimental results. Section 6 presents the conclusion.

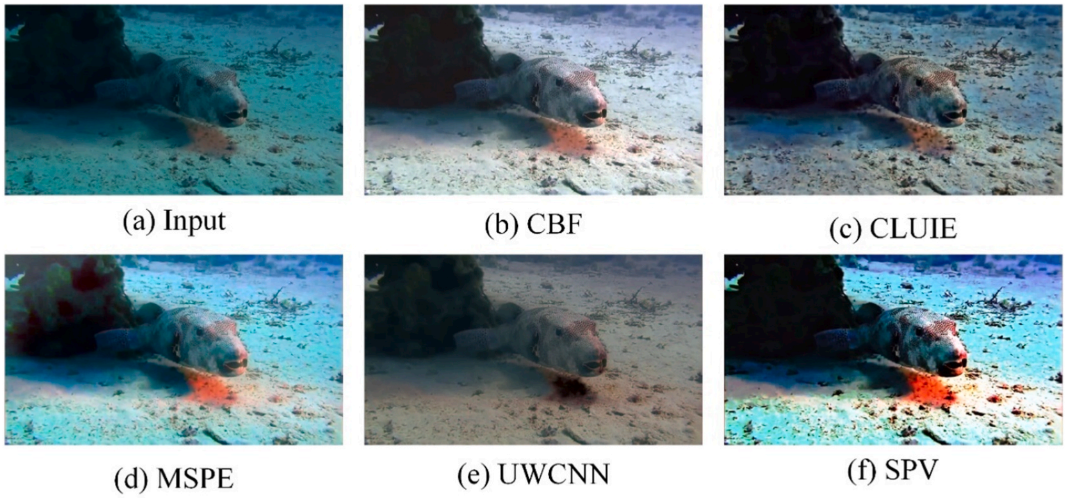

Fig. 1. The comparison of SPV with advanced non-physical model, physical model, and deep learning-based underwater image enhancement methods.

2. Related work

In recent years, there has been rapid development in the fields of underwater science and engineering research, as well as marine science research, making it particularly important to obtain clear, undegraded underwater images for research purposes. Researchers have been dedicated to developing algorithms to improve the quality of underwater images, thereby supporting more accurate and in-depth scientific exploration. These algorithms can be classified into three categories: non-physical model enhancement methods, physical model-based enhancement methods, and deep learning-based methods. Specifically, non-physical model enhancement methods directly manipulate pixel values and image structures through image processing techniques, physical model-based methods restore images by establishing physical models of underwater light propagation, deep learning-based methods learn the mapping relationship from degraded images to clear images by training deep neural networks. These methods have made considerable progress in enhancing the visual quality of underwater images, providing strong support for marine engineering.

2.1. Non-physical model enhancement methods

Traditional image processing techniques based on non-physical models are a class of techniques that use mathematical and statistical methods to enhance image quality without relying on physical models to describe the propagation characteristics of light in the medium. These methods typically directly manipulate pixel values through image processing algorithms to improve contrast, visibility, color, and details of the image. Non-physical model underwater image enhancement techniques are characterized by their simplicity, flexibility, efficiency, and wide applicability, and have been widely used in the field of underwater image enhancement. In recent years, non-physical model underwater image enhancement methods have shown significant effectiveness. For example, Zhou et al. (2023a) proposed a method based on the histogram equalization to address issues in underwater images. This method enhances features adaptively by extracting the statistical characteristics of the image to estimate the degree of feature drifts in various regions, improving the visual effects of underwater images. Zhang et al. (2022b) adopted a strategy that combines color fidelity optimization and a fusion strategy based on maximal attenuation maps to correct the color of input images and enhance local details. They used cumulative maps and squared cumulative maps to calculate the mean and variance of local image blocks, adapting the contrast of input images accordingly. Additionally, they applied a color balance strategy to balance the differences in the a and b channels in the CIELAB color space, making the image colors more consistent with human visual perception. To address issues such as white balance distortion, color cast, low visibility, and low contrast in underwater images. An et al. (2024) proposed a hybrid fusion method for underwater image enhancement. This method includes white balance adjustment, visibility recovery, and contrast enhancement, using a hybrid strategy to improve image visibility, contrast, and color cast, effectively solving degradation issues in underwater images. Li et al. (2023) developed an adaptive weighted multi-scale retinal method to enhance underwater image quality. The method divides the image into sub-blocks, calculates and combines global and local detail sparsity indices to determine the optimal scale parameters and weights. Subsequently, each sub-block undergoes Retinex processing for detail enhancement, color, and saturation correction. Finally, a gradient-domain fusion method using the structure tensor integrates the processed sub-blocks, significantly improving the quality of underwater images.

To address image degradation caused by light absorption and scattering, Zhuang et al. (2022) proposed an underwater image enhancement method based on a hyper-Laplacian reflectance priors inspired Retinex-based model. This method establishes a hyper-Laplacian reflectance prior by applying an l_1 or l_2-norm penalty on the first and second-order gradients of reflectance, using this prior to enhance prominent structures and details while restoring true colors. Additionally, it uses the l_2 norm to accurately estimate illumination, decomposing the complex problem of underwater image enhancement into independent estimations of reflectance and illumination. Each subproblem is optimized through an alternating minimization algorithm, ultimately obtaining an illumination-corrected image. Zhou et al. (2022b) classified color cast into several categories based on the ratio of average values in the RGB channels to calculate the color loss rates of RGB channels in underwater images under different scenes. They developed a multi-scene color restoration method to correct color cast. Finally, to enhance image contrast while preserving color consistency, they applied a 64-block multi-contrast factor histogram stretching technique.

However, non-physical model methods lack modeling of underwater scattering and absorption, which leads to limited accuracy in restoring color and structural details because they do not consider the physical degradation process. Without physical constraints, the enhancement often causes over-adjustment, resulting in color distortion, oversaturation, and visual artifacts. Additionally, these methods rely heavily on manual parameter tuning due to the absence of physical guidance, making them less adaptive and less efficient in practical applications. For example, Zhang et al. (2022c) proposed an underwater image color correction method that compensates for inferior color channels to address color shift phenomena. They also used a new histogram strategy to improve global and local contrast. Finally, they applied a multi-scale fusion strategy for enhancement and used a multi-scale unsharp masking strategy to further sharpen the image, enhancing visual quality. Nevertheless, this method does not explicitly model the underwater imaging process and relies on handcrafted operations without considering scattering and absorption effects. As a result, it may require manual parameter tuning to adapt to different scenes. Moreover, due to the absence of physical constraints, its performance tends to be unstable in complex water conditions and under varying scattering intensities, limiting its generalization and robustness in real-world underwater applications.

2.2. Physical model-based enhancement methods

Physical model-based enhancement methods remove haze from images by simulating the formation process of haze. This approach typically relies on the atmospheric scattering model and uses prior knowledge to estimate atmospheric light and transmission, thereby generating haze-free images with good visual effects. These methods can accurately restore image details and true colors under various concentrations and types of haze. Alenezi et al. (2022) proposed a method for estimating global background light using the gradients and eigenvalues of color channels. By calculating these features, the method accurately estimates the global background light in the image, effectively removing haze, and enhancing image clarity and contrast. Additionally, it uses the absorption differences of different color channel wavelengths to estimate the transmission map, significantly improving underwater image quality through a deep learning network. Liang et al. (2021b) proposed a systematic underwater image enhancement method, including a dehazing method guided by attenuation maps for color correction and detail preservation. This method processes color channel information using piecewise linear transformation and estimates medium transmission using the maximum intensity prior, generating images with vivid colors and fine details. To address unpredictable changes in visual characteristics in underwater environments, Chang et al. (Chang, 2020) proposed an adaptive transmission fusion method. This method introduces two different background light calculation methods and computes and fuses two transmission maps. Finally, it uses point spread function deconvolution and color compensation to generate the final scene brightness. Song et al. (2020) proposed an underwater image enhancement method combining image restoration and color correction. First, they established a manually annotated background light database, providing a statistical model for background light estimation by analyzing the relationship with the histogram distribution of underwater images. Then, using a new underwater dark channel prior and a scene depth map based on underwater light attenuation prior and adjusted reversed saturation map, they estimated and corrected the transmission map for the R channel. Finally, they estimated the transmission map for the G-B channels based on the attenuation ratio differences between the R channel and the G-B channels.

Zhou et al. (2021) proposed an image restoration method based on underwater scene feature priors. They estimated background light using smoothness, hue, and brightness to alleviate color distortion, compensated for the red channel to correct the transmission map, and refined the transmission map using a structure-guided filter. Muniraj et al. (Muniraj and Dhandapani, 2021) estimated depth through the differences in channel intensity priors. Additionally, they enhanced the contrast of estimated depth by normalizing the channel intensity priors using histogram stretching. Furthermore, they proposed a saturation correction factor for color correction, which estimates artificial background light and addresses uneven illumination in turbid images. Finally, they refined the estimated transmission depth using a filter.

However, physical model-based enhancement methods typically rely on accurately estimating key physical parameters such as transmission and background light. In complex and variable underwater environments, precise estimation of these parameters is challenging, and estimation errors often significantly affect enhancement performance. Additionally, physical models depend on identifiable image features, such as the dark channel prior and color linearity assumptions, which limits their effectiveness under severe blur, low illumination, or high noise conditions. For example, Hao et al. (2023) proposed a Laplacian variational model based on the underwater imaging model, which effectively alleviates color distortion, blurriness, and low contrast in underwater images by utilizing transmission maps and background light information. The method designs a novel fidelity term to constrain the radiance scene and introduces divergence-based regularization to enhance structural and texture details. It also employs a brightness-aware hybrid algorithm and a quadtree subdivision scheme to optimize the estimation of transmission and background light. Although this method demonstrates excellent physical consistency and image restoration accuracy, it still has several limitations. Specifically, its performance degrades in extremely low-light or turbid conditions due to its reliance on accurate transmission and background light estimation. Moreover, the method involves high computational complexity and is less effective in dynamic scenes or under non-uniform lighting, limiting its real-time applicability and generalization.

2.3. Deep learning-based enhancement methods

Deep learning-based enhancement methods utilize neural networks to learn dehazing features and patterns from many paired hazy and haze-free images. These methods typically construct end-to-end networks that input hazy images and output dehazed images. Through training, the network can learn to remove the effects of haze, restoring image clarity and detail. Deep learning dehazing methods do not require manually designed algorithms and can automatically learn features. They perform well in handling complex scenes and different types of haze, have good generalization capabilities, and can simultaneously optimize dehazing and other image enhancement tasks. In recent years, with the rapid development of deep learning dehazing methods, many deep learning-based dehazing methods have emerged. Zhang et al. (2023b) proposed an underwater image enhancement method based on generative adversarial networks, employing a new hierarchical attention aggregation with multi-resolution feature learning to address color deviation, underexposure, and blurring in underwater images. Zhang et al. (2023b) proposed a reflection-guided underwater image enhancement network, designed for extreme underwater scenarios. The network combines the complementary information of reflectance encoder and raw encoder to reduce the influence of different environmental scenes and improves feature expressiveness and robustness through multi-scale feature extraction and context-aware multi-level integration modules, avoiding artifacts and ensuring pixel precision. Qi et al. (2022) proposed a novel underwater image enhancement network that introduces semantic information through regional enhanced feature learning. This network uses a semantic region enhancement module to learn local enhanced features of multi-scale semantic regions and feeds these features into the main branch, which extracts global enhanced features at the original image scale, ultimately achieving semantically consistent and visually superior enhancement effects. Li et al. (Li and Chen, 2020) proposed an adaptive algorithm that combines randomly wired neural networks and synergistic evolution to effectively enhance underwater images. This algorithm performs color adjustment, contrast improvement, and brightness enhancement sequentially, and enhances details through edge-preserving techniques. Additionally, the authors developed a multi-strategy cooperative evolution algorithm to determine the optimal parameter values. Experimental results show that this model significantly improves the quality of underwater images both subjectively and quantitatively. Yin et al. (2024) proposed a new unsupervised underwater image enhancement framework that jointly learns water layer generation and removal through disentangled representations. Specifically, they designed a bidirectional disentanglement network, with each unidirectional network containing a cycle of water layer removal and generation, ensuring image consistency after one cycle. Additionally, they proposed a dual-stage contrastive loss to improve disentanglement capability through joint implicit constraints of first order and second-order features.

Although deep learning methods possess powerful nonlinear modeling capabilities and strong performance potential, they still face several limitations. First, such methods heavily rely on the diversity and representativeness of training data, yet it is challenging to collect datasets that comprehensively cover the wide variety of degradation types in real underwater environments. Second, most deep models function as "black boxes" and lack interpretability, making their performance unstable in previously unseen scenarios. Third, deep learning-based model training is computationally expensive and resource-intensive, making it unsuitable for real-time or resource-constrained applications. In addition, when the training data primarily consists of synthetic images, domain shift issues can significantly impair the model's generalization ability on real underwater images. For example, Guan et al. (2023) proposed an underwater image enhancement method based on a conditional denoising diffusion probabilistic model. This method leverages the strengths of diffusion models in generating high-quality images by using color-compensated degraded images as conditional guidance to restore underwater images. However, this approach relies on a large amount of annotated and preprocessed samples for conditional construction, resulting in long training cycles and high computational costs. Moreover, in scenarios with significant variations in water properties or complex lighting conditions, the method may suffer from inaccurate color restoration or insufficient detail recovery, compromising the stability and consistency of the enhancement results.

3. Proposed method

In this section, we provide a comprehensive overview of the proposed method, which comprises four main components including the spectral prior dehazing module, the color correction module, the visibility enhancement module, and the detail enhancement fusion module. The flowchart of this method is shown in Fig. 2. First, the spectral prior dehazing module achieves the dehazing effect by simulating underwater imaging principles. Next, the color correction module addresses the

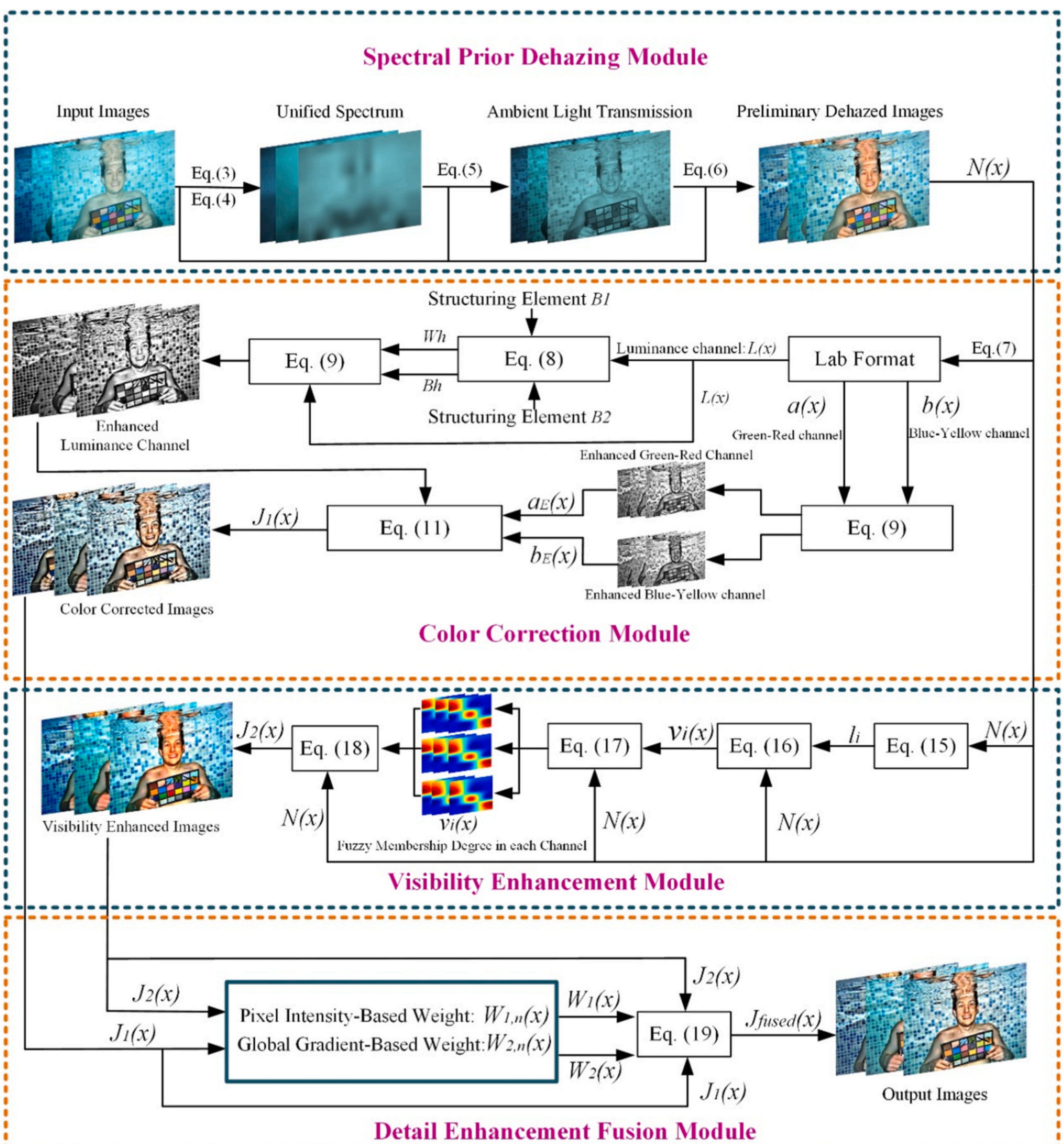

Fig. 2. Flowchart of the proposed method. In SPV, input hazy images first go through the spectral prior dehazing module. Then, the color correction and visibility enhancement modules simultaneously process the dehazed image. Finally, the image undergoes detail enhancement fusion, resulting in the final output image.

problem of color degradation caused by scattering and absorption of underwater substances. The visibility enhancement module is used to improve the overall clarity of the image. Finally, the detail enhancement fusion module integrates the outputs of the visibility enhancement and color correction modules, enabling complementary advantages between structural clarity and color fidelity, and thereby producing more visually balanced and perceptually improved enhancement results. The following sections will discuss the details and technical principles of each module. Fig. 3 illustrates the algorithm flowchart of the SPV algorithm.

3.1. Spectral prior dehazing module

Underwater images are often affected by the scattering and absorption of light, resulting in blurriness and color distortion. Methods based on physical models can simulate and correct these phenomena, thereby restoring the true radiance of underwater scenes. However, the effectiveness of physical models depends crucially on the accurate estimation of the transmission map and background light. Estimating the transmission map typically relies on certain prior knowledge, such as the dark channel prior (He et al., 2010) or the underwater dark channel prior (Liang et al., 2021c). Nevertheless, these methods rely solely on the green and blue channels to estimate the transmission and apply this estimation uniformly across all channels. When the color information among channels is inconsistent, this approach easily leads to errors in transmission estimation, which in turn affects the accuracy of color restoration. Therefore, we propose a transmission estimation method based on the unified spectrum. The unified spectrum refers to the consistent and stable spectral characteristic formed by the fixed ratio of RGB channel intensities in distant underwater regions dominated by ambient light. By utilizing this globally consistent spectrum as a reference, the method effectively reduces the uncertainty caused by red channel attenuation and complex lighting conditions, thereby achieving more accurate transmission estimation.

Typically, physical methods for underwater image dehazing are based on an imaging model that accounts for the scattering and absorption of light as it propagates through water, as shown in Eq. (1):

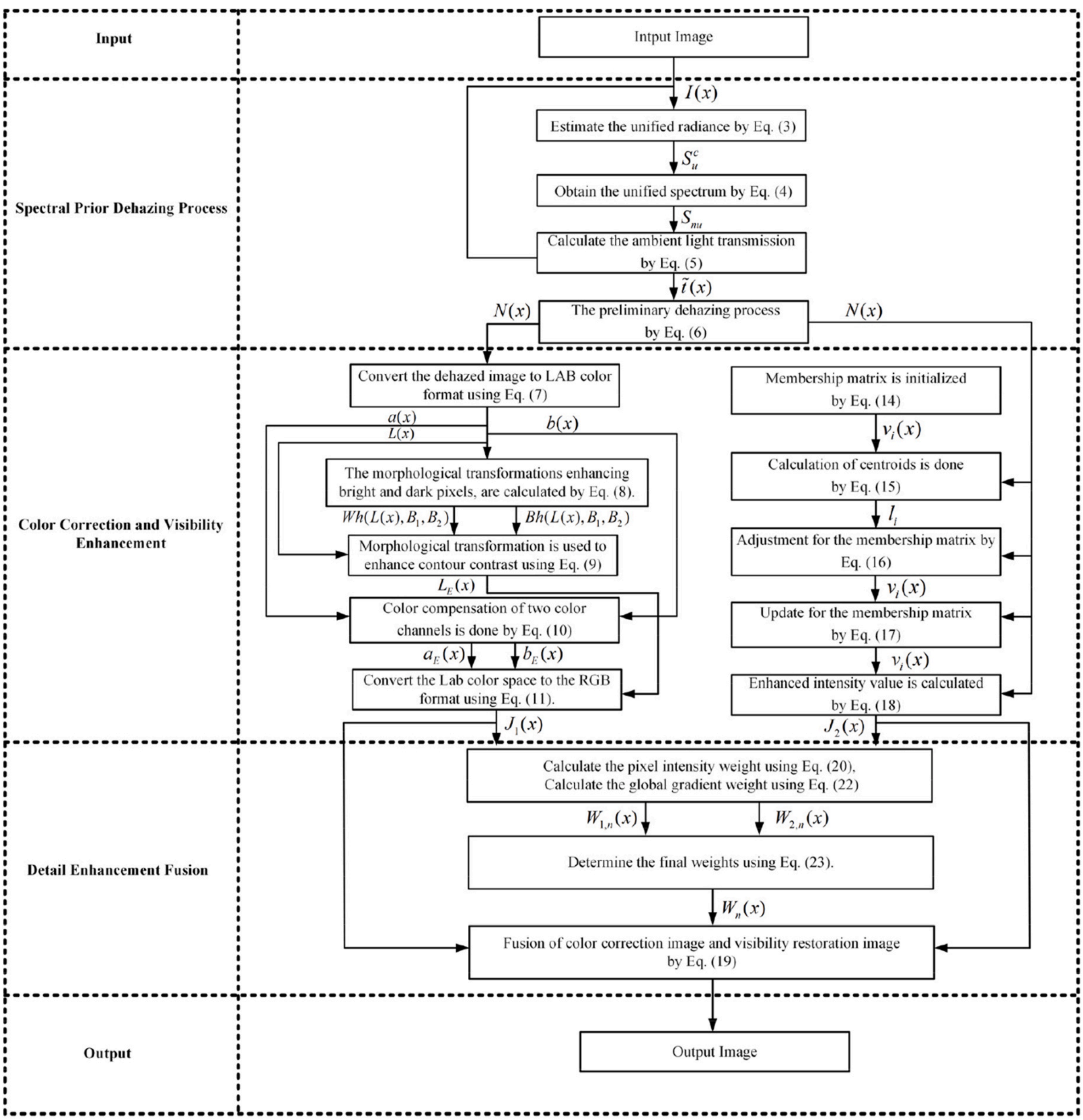

Fig. 3. SPV's algorithm flowchart. The input images are processed through several computational stages, culminating in the final enhanced output image.

$$I(x) = t(x)N(x) + \tilde{t}(x)A. \tag{1}$$

Here, $I(x)$ represents the input image, $t(x)$ denotes the transmission at pixel position x, $N(x)$ indicates the scene radiance at pixel position x, which is the image after dehazing. $\tilde{t}(x)$ is numerically equal to $1 - t(x)$, representing the transmission of the ambient light, and A represents the ambient light. Eq. (2) is used to describe the transmission, where β is the attenuation coefficient, representing the water's ability to absorb and scatter light. $d(x)$ represents the distance that light travels at pixel position x.

$$t(x) = e^{-\beta d(x)}. \tag{2}$$

in underwater images, although pixel intensities vary across different regions due to object reflectance, local illumination, and depth differences, the color ratio structure tends to exhibit a certain level of consistency in distant areas. Based on this observation, we extract the stable proportion among the RGB channels as the dominant spectral structure of ambient light, referred to as unified radiance. This feature serves as a global color reference for modeling the direction of ambient light and estimating transmission. The corresponding formulation is given as follows:

$$S_u^c = \frac{1}{|\Omega|}\sum_{x\in\Omega} I^c(x), c \in \{R, G, B\}. \tag{3}$$

The unified radiance S_u^c represents the spectral characteristics of a specific color channel. Here, the region Ω refers to the set of all pixels in the entire image, and $|\Omega|$ denotes the total number of pixels in this region. $I^c(x)$ is the intensity value of color channel c at position x. The spectral characteristics of the ambient light are estimated by averaging the intensity values of all pixels across the entire image for each color channel. Subsequently, by dividing the unified radiance S_u by its L_1 norm, we normalize it to obtain the unified spectrum S_{nu}, which has the property that the sum of its elements is equal to 1, as shown below:

$$S_{nu} = \frac{S_u}{\| S_u \|_1}. \tag{4}$$

Next, the unified spectrum S_{nu} is used to estimate the transmission. The observed intensity $I(x)$ is projected onto the normalized unified spectrum S_{nu}, and the resulting projection value indicates the strength of the color component that aligns with the direction of ambient light. This value reflects the relative contribution of ambient illumination at pixel x, and serves as the basis for estimating the transmission. A projection value closer to 1 suggests that the observed color is more consistent with the ambient light direction, indicating a stronger influence of ambient light. In such cases, $\tilde{t}(x)$ approaches 1, and the transmission $t(x)$ tends toward 0, the equation is defined as follows:

$$\tilde{t}(x) = \langle I(x), S_{nu} \rangle \cdot S_{nu}. \tag{5}$$

Finally, the scene radiance $N(x)$, which is the result after dehazing, is estimated using Eq. (6). Here, α is a relaxation parameter with a value range between $(0, 1]$, used to adjust the influence of the transmission rate. t_0 is a lower bound used for stabilization in the calculation to prevent the denominator from approaching zero, in this paper, it is set to 0.001. The ambient light A is estimated by taking the mean of the top 0.1% of pixels with the highest $\tilde{t}(x)$ values. After applying the dehazing algorithm, the enhanced result $N(x)$ was obtained:

$$N(x) = \frac{I(x) - \alpha\tilde{t}(x)A}{\max(1 - \alpha\tilde{t}(x), t_0)}. \tag{6}$$

3.2. Color correction module

In underwater environments, substances cause the absorption and scattering of light, leading to a series of color degradation phenomena. Therefore, color correction of underwater images is necessary. However, traditional Gray-World algorithm (Buchsbaum, 1980) and white balance algorithms fail to consider the visual perception characteristics of the human eye during color adjustment, which may result in color correction outcomes that are inconsistent with human visual perception. In contrast, the CIELAB color model is more aligned with human color perception, providing more natural and realistic effects in color adjustment and correction. Based on this, we proposed a color correction method using the CIELAB color space (Zhang and Wandell, 1996). First, the RGB format input image $N(x)$ is converted to the LAB format image $N_{\mathrm{Lab}}(x)$ as shown in Eq. (7):

$$N_{\mathrm{Lab}}(x) = f_{\mathrm{Lab}}(N(x)). \tag{7}$$

In the LAB format, $N_{\mathrm{Lab}}(x)$ includes three channels: $L(x)$, $a(x)$, and $b(x)$.

To enhance the representation of image details, we apply a multiply contour bougie morphology transform on the L channel, which represents luminance information. This technique effectively restores and highlights the contours of objects in the image, making the image details clearer and more vivid, thereby improving the overall visual quality. Eq. (8) and Eq. (9) described this transformation process:

$$\begin{cases} Wh(L(x), B_1, B_2) = L(x) - L(x)((L(x)\circ B_1)\cdot B_2) \\ Bh(L(x), B_1, B_2) = L(x)((L(x)\cdot B_1)\circ B_2) - L(x) \end{cases}, \tag{8}$$

$$L_E(x) = L(x) + \omega\sum_{i=1}^{n}(Wh_i(L(x), B_1, B_2) - Bh_i(L(x), B_1, B_2)). \tag{9}$$

First, we enhance the luminance channel $L(x)$. In Eq. (8), *Wh* is used to emphasize the difference between the foreground and background of the image, increasing the luminance of the target by maintaining pixels that are brighter than their surroundings. Conversely, *Bh* performs the opposite operation of *Wh*, enhancing the dark pixels. In the equations, $\circ$ represents the morphological opening operation, and $\cdot$ represents the morphological closing operation. B_1 and B_2 are structuring elements, with B_1 being a square structuring element with a side length of 15, is used to suppress small bright regions and emphasize relatively large-scale background structures through morphological opening and closing, and B_2 being a square structuring element with a side length of 5, focuses on finer structures and enhances local contrast. This multi-scale design helps distinguish salient foreground details from background variations, improving the overall luminance enhancement performance. Next, to further highlight the foreground and enhance the contours of objects, we subtract the bright pixels from the dark pixels and perform multiple transformations, as shown in Eq. (9). Here, the number of iterations is set to $n = 10$ and the parameter $\omega = 0.1$ is used to adjust the extraction amount. These parameters are empirically determined to achieve a good balance between enhancement quality and computational cost. As shown in Fig. 4, when $n = 10$ and $\omega = 0.1$, the image exhibits the clearest edges and textures, indicating better enhancement performance.

To compensate for the color shift issue in underwater images, we proposed a color compensation algorithm. This algorithm utilizes a salient radiation region mask and Gaussian filtering, achieving detail enhancement and overall color correction of the image through multiple iterative transformations. Eq. (10) is as follows:

$$\begin{cases} a_E(x) = a(x) - \kappa\cdot M(x)\cdot G_a(x), \\ b_E(x) = b(x) - \lambda\cdot M(x)\cdot G_b(x). \end{cases} \tag{10}$$

Here, $a_E(x)$ and $b_E(x)$ represent the color-compensated values of the a and b channels, respectively. In the formula, $a(x)$ and $b(x)$ are the original image's a and b channel values. The compensation process is achieved by subtracting a compensation term, which is determined by the adjustment coefficients κ and λ, the mask operation $M(x)$, and the products of the Gaussian filtering operations $G_a(x)$ and $G_b(x)$. The mask operation $M(x)$ is used to retain significant radiation in the source position, while the Gaussian filter G smooths the mask values, making the color-compensated image more natural and realistic. For underwater

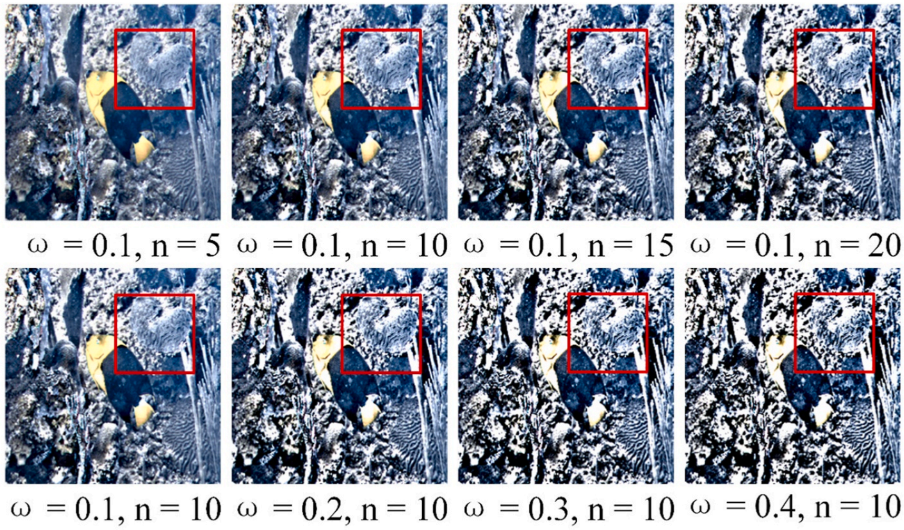

Fig. 4. Comparison results generated by different enhancement parameters n and ω. When n < 10, enhancement is limited and fails to reveal global structures. When n > 10, more structural details are enhanced, but excessive values may introduce artifacts. A small ω yields subtle, natural enhancement, while a large ω improves contrast but may cause over-enhancement. when n = 10 and ω = 0.1, the image exhibits the clearest edges and textures, indicating better enhancement performance.

images, we set κ to 1 and λ to 0.7 (Ancuti et al., 2019). After performing the color corrections in the CIELAB space, the corrected image is then converted back to the RGB format as shown in Eq. (11):

$$J_1(x) = f_{\mathrm{RGB}}(N_{\mathrm{Lab}}(x)). \tag{11}$$

3.3. Visibility enhancement module

The complex optical environment underwater often leads to low-light images characterized by insufficient brightness, low contrast, and blurred details, especially with indistinct transitions and unclear boundaries between bright and dark regions. Traditional image enhancement methods frequently encounter issues such as over-enhancement, loss of detail, and artifact generation when processing these images, making it difficult to balance enhancement effectiveness with visual naturalness. To address these challenges, this paper proposes a visibility enhancement module based on fuzzy c-means clustering. The soft clustering property of this module allows each pixel to belong to multiple clusters with varying degrees of membership, better accommodating the smooth luminance transitions typical in underwater low-light images. This approach achieves smoother and more natural enhancement, avoiding the edge discontinuities and detail breakage caused by traditional hard clustering methods. Meanwhile, fuzzy c-means clustering enhancement module effectively preserves image structure and color information, reduces artifacts and color distortions, and improves the visual quality and overall image fidelity. In this section, we elaborate on the implementation process of the visibility enhancement module.

For the preliminary dehazed image $N(x)$, we first separate it into the R, G and B color channels. Then, we initialize the parameters, setting the fuzzification degree $m = 2$ and the number of clusters $c = 3$. The image is then subjected to the fuzzification process as shown in Eq. (12):

$$L = U\{v(N(x))\} = \left\{\frac{v(x)}{N(x)}\right\}. \tag{12}$$

Here, L represents the result obtained through fuzzy set operations, $v(x)$ represents the membership degree of a certain feature for pixel $N(x)$ which is the pixel intensity level at position x. For a color image, it can be represented as the union of the color channels, as shown in the following equation:

$$L = \cup\{L_R, L_G, L_B\}. \tag{13}$$

Here, L_R, L_G and L_B represent the fuzzified images of the red, green, and blue color channels, respectively.

Next, the membership degrees of intensity levels are calculated using the fuzzy c-means clustering algorithm. The membership matrix is initialized as shown in Eq. (14), where c represents the number of clusters, and $v_i(x)$ is an element in the membership matrix, indicating the degree to which the data point x belongs to the $i-th$ cluster. The calculation process for the centroids is shown in Eq. (15), where the cluster center l_i in the fuzzy c-means clustering algorithm is obtained by a weighted sum of each data point x's membership degree $v_i(x)$ and the fuzzification coefficient m, and then normalized. Here, n represents the total number of pixels in the image. The denominator is the sum of all membership degrees $v_i(x)$. The update process for the membership matrix $v_i(x)$ in the fuzzy c-means clustering algorithm is shown in Eq. (16). It represents the relative relationship between the distance of data point x to cluster i and its distance to other clusters k, using the squared Euclidean distance. These steps are repeated for several iterations until the maximum number of iterations is reached.

$$\sum_{i=1}^{c} v_i(x) = 1, \tag{14}$$

$$l_i = \frac{\sum_{x=1}^{n} v_i^m(x)\cdot N(x)}{\sum_{x=1}^{n} v_i^m(x)}, 1 \le i \le c, \tag{15}$$

$$v_i(x) = \sum_{k=1}^{c}\left(\frac{\|N(x) - l_i\|^2}{\|N(x) - l_k\|^2}\right)^{-\frac{1}{m-1}}. \tag{16}$$

After this, the membership degrees are updated as shown in Eq. (17), where x represents the pixel position in the image, $N(x)$ represents the intensity value of the pixel, $v_i(x)$ is the fuzzy membership degree of the x position in the $i-th$ cluster, and l_i is the centroid of the $i-th$ cluster. The $v_i(x)$ is adjusted based on the relationship between $N(x)$ and l_i.

$$v_i(x) = \begin{cases} v_i(x), & N(x) \le l_i, i = 1,2,3 \\ 1 + (1 - v_i(x)), & N(x) > l_i, i = 1,2,3 \end{cases}. \tag{17}$$

Finally, the enhanced intensity value $J_2(x)$ is calculated by taking the weighted average of the modified membership degrees for each intensity level across all clusters. In Eq. (18), $v_i(x)$ represents the updated membership degree, and $N(x)$ is the intensity corresponding to the position x.

$$J_2(x) = \sum_{i=1}^{c} \frac{\nu_i(x) \cdot N(x)}{c}. \tag{18}$$

3.4. Detail enhancement fusion module

The fusion strategy combines the strengths of different images by preserving more useful information through merging data from multiple sources. For example, visibility-restored images perform better under low-light or hazy conditions while contrast-enhanced images excel in capturing fine details and edge clarity. By integrating these complementary features, a richer and more detailed image can be produced. To achieve this, we propose a fusion strategy that leverages both pixel intensity and global gradient information to dynamically assign weights to each input image.

Compared with CNN-based fusion methods, our approach offers key advantages in being training-free and computationally efficient, avoiding reliance on large-scale labeled datasets and high-performance hardware. This makes it especially suitable for real-world underwater scenarios where image variability is high and training data are scarce. Moreover, CNN-based methods may introduce artifacts or distortions when training is insufficient or when applied across different domains. In contrast, our fusion module's weighting mechanism based on pixel intensity and global gradient effectively preserves brightness from low-light enhanced images and structural details from dehazed images, resulting in a clearer and more natural fused output without visual artifacts.

The proposed fusion algorithm is shown in Eq. (19). In this formula, $J_{fused}(x)$ represents the pixel value at position x in the fused image. The fusion process involves combining the pixel values of two input images $J_1(x)$ and $J_2(x)$. The weight $W_1(x)$ and $W_2(x)$ for each input image at the corresponding pixel position is dynamically adjusted based on the pixel intensity and global gradient of the image, to optimize the pixel performance of each image. The goal of this fusion strategy is to create a visually optimal image by selecting the best-performing pixels from each input image at different positions. This method can significantly enhance the clarity and detail performance of the final image, achieving superior visual effects. In this section, the specific calculation and implementation process of the weight W will be explained.

$$J_{fused}(x) = W_1(x)J_1(x) + W_2(x)J_2(x). \tag{19}$$

1) Weight Determined by Pixel Intensity

We use the weight based on pixel intensity as the first sub-weight of $W_{1,n}(x)$, aiming to improve the issue of uneven image exposure. As shown in Eq. (20), this equation is used to calculate a sub-weight $W_{1,n}(x)$ for image fusion, where n denotes the $n-th$ image, and x is the pixel positions. The equation includes an exponential function exp with a squared normalized difference as its exponent. Specifically, $J_n(x)$ represents the pixel intensity at position x in the $n-th$ image, m_n is the average pixel intensity of the image, and σ_n is the standard deviation used to adjust the influence of the difference. The difference $J_n(x)-(1-m_n)$ indicates the deviation of the pixel intensity from the average intensity of the image. When m_n is large, indicating that the overall brightness of the image is high, the equation assigns greater weight to the dark pixels. Conversely, when m_n is small, indicating that the overall brightness of the image is low, the bright pixels receive greater weight. This design effectively addresses the problem of uneven image exposure, ensuring that the details of each region are appropriately emphasized and preserved during the fusion process. The value of σ_n is given inEq. (21), and in all experiments in this paper, the value of α is set to 0.75.

$$W_{1,n}(x) = \exp\left(-\frac{(J_n(x) - (1-m_n))^2}{2\sigma_n^2}\right), \tag{20}$$

$$\sigma_n = \begin{cases} 2\alpha(m_{n+1} - m_n) & n = 1 \\ \alpha(m_{n+1} - m_{n-1}) & 1 < n < N \\ 2\alpha(m_n - m_{n-1}) & n = N. \end{cases} \tag{21}$$

2) Weight Determined by Global Gradient

We use the global gradient weight as the second sub-weight of $W_{2,n}(x)$ to improve the contrast of the image. For images with different exposure levels, regions with a slowly changing cumulative histogram. For instance, small gradients usually indicate that these areas have moderate contrast, displaying a balanced distribution of light and color. This makes the image details more prominent and clearer. During the process of image fusion or enhancement, giving greater weight to these regions can help better preserve and enhance the detail information. The equation is as follows:

$$W_{2,n}(x) = \frac{\mathrm{Grad}_n(J_n(x))^{-1}}{\sum_{n=1}^{N} \mathrm{Grad}_n(J_n(x))^{-1} + \epsilon}. \tag{22}$$

Here, $\mathrm{Grad}_n(J_n(x))$ represents the gradient of the cumulative histogram at intensity $J_n(x)$. For regions with small gradients, we take the reciprocal to assign greater weight. ϵ is a very small positive number used to prevent the denominator from becoming zero. Finally, Eq. (23) shows how to calculate the final weight $W_n(x)$ by combining the two sub-weights $W_{1,n}(x)$ and $W_{2,n}(x)$. Then, $W_n(x)$ is substituted into Eq. (19) and combined with the pyramid image decomposition method (Mertens et al., 2009) to synthesize the result $J_{fused}(x)$.

$$W_n(x) = \frac{W_{1,n}(x) \times W_{2,n}(x)}{\sum_{n=1}^{N} W_{1,n}(x) \times W_{2,n}(x) + \epsilon}. \tag{23}$$

4. Experiments

In this section, we provide a detailed description of the experiment settings used to evaluate the proposed algorithm. These settings include no-reference underwater datasets, comparative algorithms, no-reference image quality assessment metrics, and both qualitative and quantitative experiments on various datasets. These settings offer non-numerical descriptive assessments as well as objective numerical evaluations of the SPV algorithm's performance and effectiveness. We also conducted ablation studies and application tests. The experimental results indicate that, in both qualitative and quantitative comparative experiments, the SPV algorithm outperforms existing state-of-the-art methods in terms of efficiency and stability. We also conducted an efficiency analysis of the algorithm, as shown in Table 3, which further validates the advanced performance of SPV in terms of processing speed and resource utilization. The ablation studies demonstrate the effectiveness of each module of the SPV algorithm and their contributions to the overall image enhancement algorithm. Application tests based on advanced visual tasks further confirm the SPV algorithm's effectiveness and robustness in real-world scenarios, especially in edge detection, SIFT feature point matching, and geometric transformation estimation. These tests highlight the superiority of the proposed method in these applications. Overall, our experimental setup is comprehensive and reliable, and the results demonstrate the SPV algorithm's effectiveness, robustness, and adaptability. In summary, these experimental results fully validate the promising performance of the proposed SPV algorithm in various aspects, laying a solid foundation for its widespread application in practical scenarios.

4.1. Experiment settings

In this section, we provide a detailed description of the experiment settings. First, we introduce four no-reference image datasets; then, we detail five no-reference evaluation metrics. Finally, we discuss 14 of the most representative and state-of-the-art underwater image enhancement methods.

No-Reference Image Datasets: No-reference underwater image datasets refer to collections of images that do not include original clear images, used for evaluating and developing underwater image enhancement algorithms. They typically contain photos from natural underwater environments, characterized by color distortion, low contrast, and blurriness. Here, we introduce four no-reference image datasets: Color-Check7 (Ancuti et al., 2017a), Test-C60 (Li et al., 2019a), OceanDark (Marques and Albu, 2020), and Test-U45 (Li et al., 2019b).

1) Color-Check7: This dataset is an underwater image dataset based on the Macbeth Color Checker standard, containing 7 images taken by different cameras. It is used for researching and developing color correction algorithms to address color distortion caused by light absorption and scattering in underwater environments. The dataset includes standard color card images taken underwater, with each camera having different imaging characteristics. This dataset is used to evaluate and validate the performance of color correction algorithms across different devices and environments, ensuring the accuracy and authenticity of color restoration.
2) Test-C60: This dataset contains 60 challenging underwater images that lack satisfactory references due to the complex shooting environments. The Test-C60 dataset is widely used by researchers to evaluate and develop underwater image enhancement algorithms, testing the effectiveness of these algorithms in real-world applications and verifying their performance in complex underwater environments.
3) OceanDark: This dataset consists of 183 underwater images with a resolution of 1280 by 720 pixels, taken by deep-sea cameras under artificial lighting conditions. Each image was taken under

suboptimal lighting conditions, meaning the illumination was poor, resulting in varying degrees of low-light areas. This dataset aims to help researchers develop and evaluate the effectiveness and robustness of underwater image enhancement algorithms by providing real-world examples of low-light environments.

4) Test-U45: Test-U45 is a no-reference public underwater dataset that contains 45 images exhibiting three common degradation phenomena in underwater environments: color cast, low contrast, and haze effects. It provides an ideal platform for evaluating image enhancement algorithms. Researchers widely use this dataset to assess the effectiveness and robustness of underwater image enhancement algorithms, testing the algorithms' performance under different degradation conditions to verify their potential and reliability in real-world applications.

No-reference Image Quality Assessment Metrics: No-reference underwater image quality evaluation metrics assess the quality of an image by analyzing its statistical properties, color, contrast, and sharpness without needing a reference image. These metrics are designed based on the human visual system and can better reflect the subjective perception of underwater image quality by humans. In this section, we use five no-reference underwater image quality evaluation metrics: information entropy (IE) (Núñez et al., 1996), variance function evaluation (VAR) (Fletcher, 1970; Wang et al., 2018), underwater color image quality evaluation (UCIQE) (Yang and Sowmya, 2015), fog aware density evaluator (FADE) (Choi et al., 2015), and contrast enhancement-based image quality (CEIQ) (Yan et al., 2019). A high IE indicates that the grayscale distribution of the image is more complex and the image contains richer details, reflecting a greater amount of information. VAR is used to evaluate image sharpness, with a larger value indicating a clearer image. A larger UCIQE value suggests better performance in color, contrast, and saturation, indicating higher image quality. A lower FADE scores indicate less haze in the image, which means the image is clearer. A higher CEIQ value typically indicates better contrast enhancement, more detail preservation, and overall higher image quality.

Compared Methods: In the comparative experiments, we evaluated 14 state-of-the-art underwater image enhancement methods on four no-reference underwater image datasets. These methods include four non-physical model methods (CBF (Ancuti et al., 2017b), EUIVF (Ancuti et al., 2012), TSA (Fu et al., 2017), and MPSE (Li et al., 2022b)), five physical model-based methods (WCD (Chiang and Chen, 2011b), UTV (Hou et al., 2020), UDCP (Drews et al., 2013), ROP (Liu et al., 2021), and DNIM (An et al., 2025)), and five deep learning-based methods (DEA (Pan et al., 2019), UT (Peng et al., 2023), CLUIE (Li et al., 2022b), FUnIE-GAN (Islam et al., 2020), and UWCNN (Li et al., 2020)). These enhancement methods are widely used in underwater image processing. Through these comparative experiments, we effectively measured the performance of the proposed method.

4.2. Qualitative and Quantitative Comparisons on the Color-Check7 dataset

1) Qualitative Comparisons

We conducted qualitative comparison experiments on the Color-Check7 dataset to evaluate the performance of various algorithms in terms of color correction, brightness, contrast, and detail representation. As shown in Fig. 5, methods such as CBF (b) and WCD (e) moderately enhance brightness but still retain a cool tone, with color deviations not effectively corrected. EUIVF (c), ROP (k), CLUIE (l), and MSPE (j) perform notably well by significantly improving image contrast, correcting skin tones and background color casts, and enhancing image clarity. DEA (d) and DNIM (m) show partial improvement in detail visibility, though their color rendering remains slightly bluish or unnatural. UTV (f) and UWCNN (o) suffer from severe color distortion and artifacts, resulting in a noticeable decline in image quality. While UDCP (g) and FUnIE-GAN (n) exhibit high contrast. UT (h) produces an overly dark image with significant detail loss. TSA (i) improves brightness and contrast but presents noticeable uneven illumination. In contrast, SPV (p) demonstrates superior visual performance by enhancing brightness and detail while maintaining natural color fidelity.

2) Quantitative Comparisons

We systematically evaluated our method on the Color-Check7 dataset to validate the color correction performance of the proposed image enhancement algorithm across different devices and environments. Through this experiment, we comprehensively assessed the performance of underwater image enhancement algorithms in diverse application scenarios and color correction tasks. The results in Table 1 show that our method performs outstandingly across all evaluation metrics, significantly surpassing other comparative algorithms, demonstrating its superior performance and robustness in color correction.

4.3. Qualitative and Quantitative Comparisons on the Test-C60 dataset

1) Qualitative Comparisons

We conducted qualitative comparison experiments on the Test-C60 dataset to evaluate the performance of various algorithms in terms of color correction, contrast enhancement, visibility improvement, and enhancement under low light conditions in complex underwater environments. These experiments allowed us to more clearly assess the

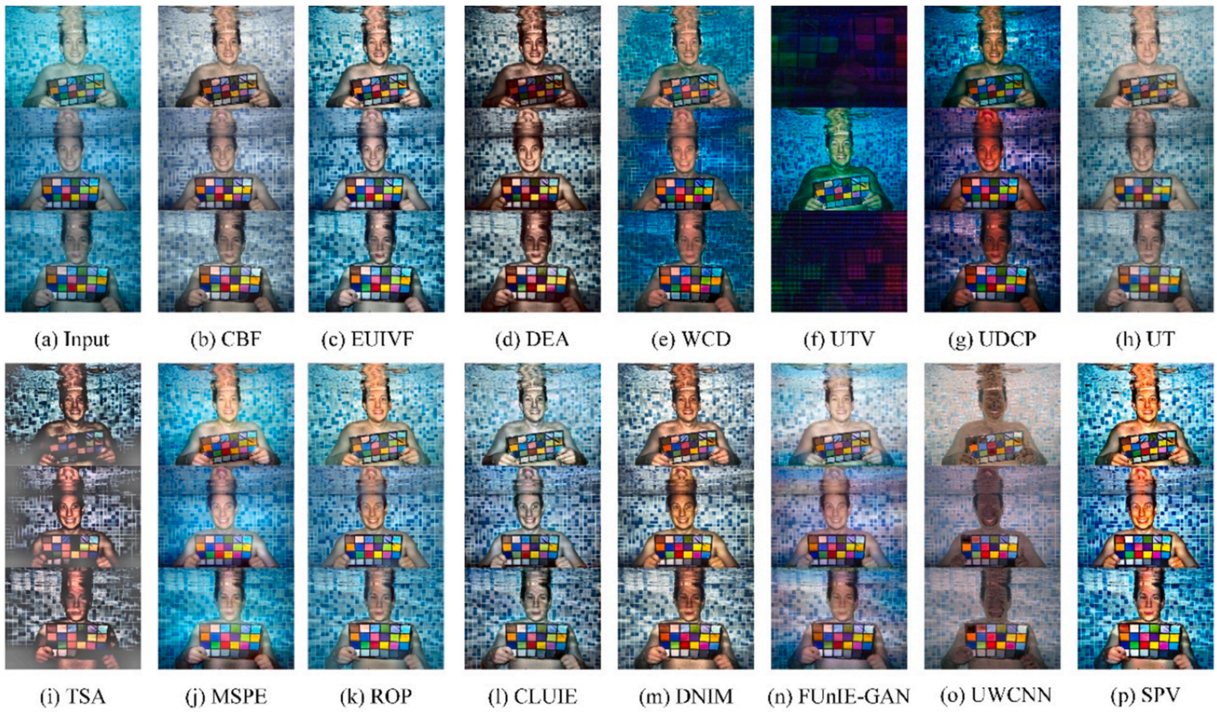

Fig. 5. Comparison of enhancement effects between the SPV algorithm and various underwater image enhancement algorithms on the Color-Check7 dataset.

Table 1
Comparative analysis of various methods on the Color-Check7 and Test-C60 dataset utilizing no-reference evaluation metrics. The best result is in bold, and the second-best result is underlined. '-' indicates the results not available.

Methods	Color-Check7					Test-C60				
	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑
CBF	7.5664	9.3570	29.8803	0.4682	3.4582	7.1439	11.1809	28.3025	1.3193	3.2680
EUIVF	7.7301	11.4686	31.6475	0.3019	3.5855	7.4607	12.1003	<u>30.7293</u>	0.9010	3.4313
DEA	7.7130	<u>14.2131</u>	29.2375	0.3481	3.5580	<u>7.5460</u>	<u>13.3355</u>	26.3345	1.1816	**3.4889**
WCD	7.0801	4.7764	28.0792	0.2459	3.1966	6.2819	3.6337	23.8348	0.9716	2.7335
UTV	4.4476	4.6408	14.8634	0.1685	2.1301	4.4638	3.0590	15.6343	–	2.1362
UDCP	7.1302	6.1009	29.1484	<u>0.2444</u>	3.1799	5.9289	4.1788	26.0029	<u>0.6734</u>	2.4961
UT	7.2470	5.6782	25.1038	0.2920	3.2415	7.0094	7.3834	24.1904	**0.6631**	3.1443
TSA	7.5174	9.4858	28.4229	0.6006	3.4599	6.4897	5.5411	22.6250	1.6705	2.8901
MSPE	7.4826	7.7908	30.1383	0.3767	3.4253	7.4402	9.3175	29.6180	0.9071	3.4214
ROP	7.4171	7.6094	31.3589	0.3509	3.3778	7.0595	8.4856	30.4021	0.9161	3.2039
CLUIE	7.6604	10.6643	30.6464	0.3081	3.5425	6.9394	7.8595	24.7935	0.9353	3.1194
DNIM	<u>7.8048</u>	13.4115	<u>32.8126</u>	0.3110	<u>3.6348</u>	**7.5581**	10.2238	26.3188	1.4757	<u>3.4865</u>
FUnIE-GAN	7.4146	7.5869	28.5883	0.4034	3.3783	6.6353	5.0215	22.4368	0.7040	2.9022
UWCNN	7.2642	6.5444	27.1922	0.5139	3.3190	6.5543	5.0574	19.6331	1.3085	2.9208
SPV	**7.8589**	**15.9903**	**36.1458**	**0.2336**	**3.6642**	7.2547	**15.8880**	**35.3324**	0.6809	3.2595

effectiveness and advantages of each algorithm in practical application scenarios. As shown in Fig. 6, the images processed by WCD (c) often exhibit severe bluish-purple color casts, while TSA (i) tends to produce unnatural cyan and greenish hues. Although UDCP (g) suffers low luminance, residual bluish-green tints persist in certain scenes. CBF (b) and CLUIE (l) demonstrate limited effectiveness in enhancing contrast and brightness, resulting in images that remain dull and lack depth. DNIM (m) provides insufficient brightness enhancement under low-light conditions, making details difficult to discern. UTV (f) causes the entire image to appear excessively dark, significantly degrading image quality. While DEA (d) and MSPE (j) have certain abilities to restore details, they still fall short in complex regions, and the sharpening process in UT (h) sometimes causes halo artifacts, affecting naturalness. Due to limitations in training data, methods

FunIE-GAN (n), and UWCNN (o) show inadequate generalization across diverse underwater scenes, producing enhancement results with obvious artificial artifacts and exhibiting a dark luminance. In contrast, SPV (p) significantly improves overall image brightness, effectively enhances detail visibility in low-light environments, and accurately restores and amplifies the inherent colors of underwater objects.

2) Quantitative Comparisons

We systematically evaluated our method on the Test-C60 dataset to validate its effectiveness and robustness in complex shooting environments. As shown in Table 1, our algorithm achieved the best results in two out of five no-reference image evaluation metrics and secured advanced performance in the other metric. This indicates that our algorithm exhibits excellent performance and robustness in handling images taken in complex lighting and underwater environments, effectively addressing the challenges of image processing in various complex conditions.

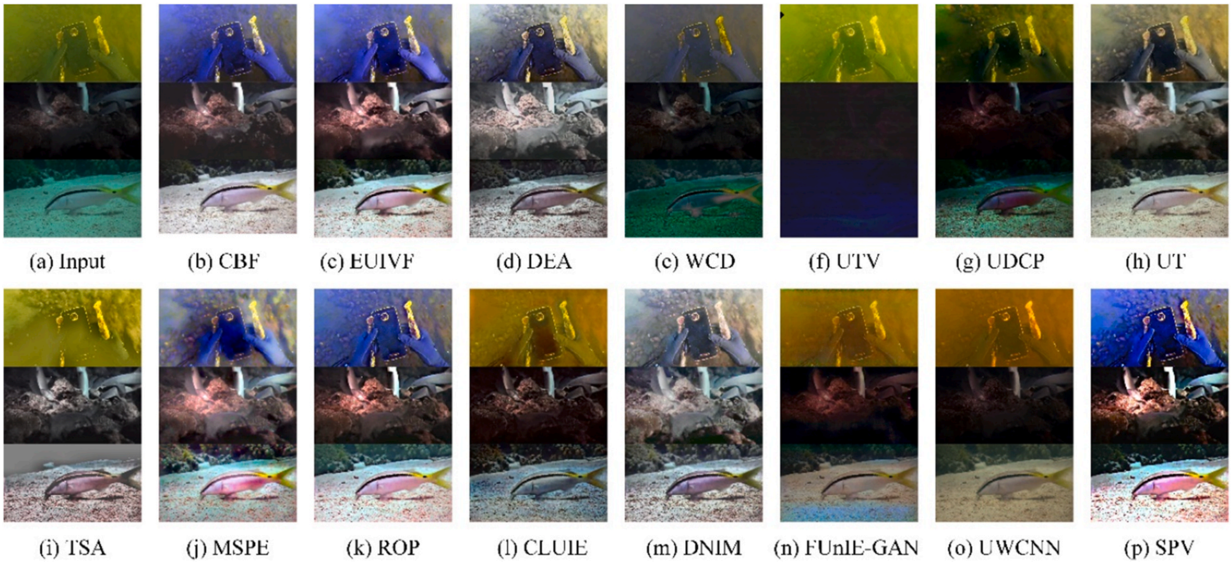

Fig. 6. Comparison of enhancement effects between the SPV algorithm and various underwater image enhancement algorithms on the Test-C60 dataset.

4.4. Qualitative and Quantitative Comparisons on the OceanDark dataset

1) Qualitative Comparisons

We conducted qualitative comparison experiments on the OceanDark dataset to evaluate the performance of various algorithms in terms of color correction, contrast enhancement, and visibility improvement in poorly lit and complex underwater environments. The experimental results in Fig. 7, CBF (b) and EUTVF (c) achieve some improvement in contrast and clarity but still suffer from inaccurate color restoration, exhibiting noticeable color casts that affect the natural appearance. UTV (f) and UDCP (g) show uneven dehazing results and tend to over-correct colors, leading to color distortion and severe color shifts. DEA (d) and UT (h) perform poorly in enhancing contrast and color recovery, resulting in images that appear dull and lack vividness and depth. WCD (e) causes local underexposure, leading to insufficient overall depth and poor detail preservation. TSA's (i) dehazing is incomplete, leaving images blurry with significant detail loss. MSPE (j) produces obvious edge artifacts and inadequate color correction, which degrade the overall visual quality. ROP (k) introduces greenish and yellowish color casts along with haze effects. CLUIE results in color bias and uneven brightness distribution, impairing the overall visual effect. DNIM (m) suffers from low contrast, producing flat and weak images, while FunIE-GAN (n) and UWCNN (o) exhibit noticeable reddish tones, unnatural texture details, and struggle to adapt to complex and varied underwater environments. In contrast, the SPV (p) algorithm shows significant advantages in image processing. It better preserves image details, especially under low-light conditions, while effectively correcting colors and eliminating color bias. This demonstrates that the SPV algorithm excels in image correction capabilities in complex underwater scenarios.

2) Quantitative Comparisons

We systematically evaluated our method on the OceanDark dataset to comprehensively validate the algorithm's image enhancement performance under low-light conditions. Through this experiment, we aimed to assess the algorithm's ability to improve image quality in extreme lighting conditions. According to the data in Table 2, the SPV algorithm's image enhancement effects under low-light conditions are particularly notable. Specifically, the SPV method outperforms other

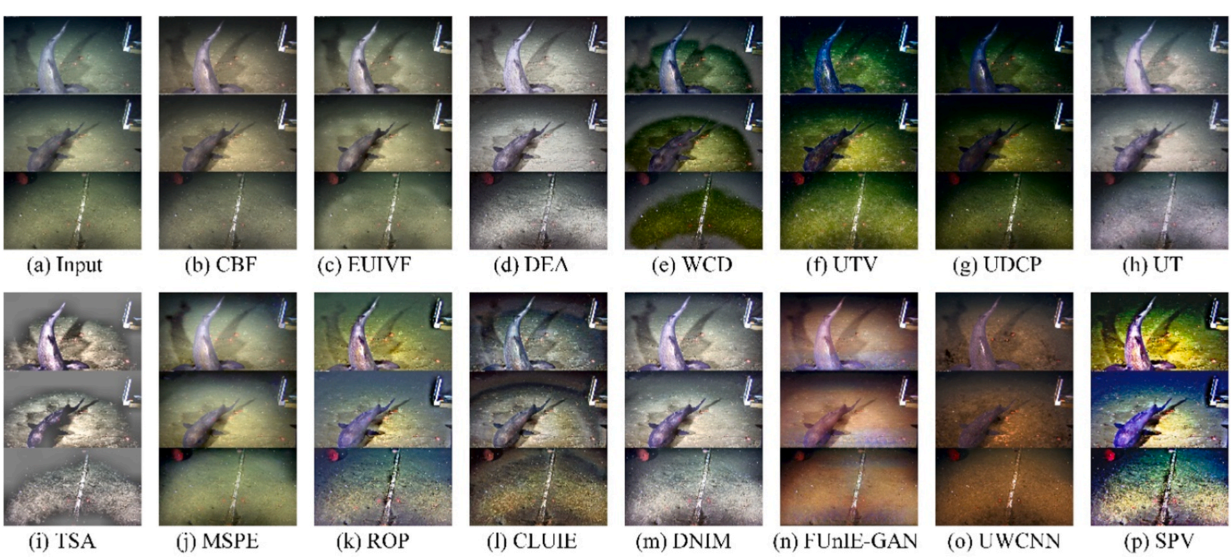

Fig. 7. Comparison of enhancement effects between the SPV algorithm and various underwater image enhancement algorithms on the OceanDark dataset.

Table 2
Comparative analysis of various methods on the OceanDark and Test-U45 datasets utilizing no-reference evaluation metrics. '-' indicates the results not available. The best result is in bold, and the second-best result is underlined.

Methods	OceanDark					Test-U45				
	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑
CBF	7.2512	12.2787	26.3899	1.6212	3.3597	7.5176	12.2252	29.6116	0.3903	3.4268
EUIVF	7.4923	12.9737	28.3134	1.3951	3.4611	7.6427	12.9483	<u>31.5868</u>	0.2689	3.5165
DEA	7.5483	12.1488	24.3546	1.5893	3.4913	7.6483	<u>14.8515</u>	27.4930	0.3344	3.5087
WCD	6.9593	8.7879	30.5204	0.7069	3.1177	6.6934	4.1033	25.7559	0.3255	2.8964
UTV	6.9662	11.7943	<u>32.0164</u>	**0.4032**	3.0626	5.0057	5.7644	28.4896	–	2.3554
UDCP	6.6234	9.3267	30.2752	<u>0.6041</u>	2.7710	7.0427	7.2425	30.9324	<u>0.2355</u>	3.1115
UT	7.4834	11.7608	25.4409	0.7777	3.4350	7.2756	7.9210	25.1630	0.4386	3.2712
TSA	6.2183	7.1888	23.4129	1.8166	2.8715	6.8059	5.4926	22.8243	0.5735	2.9993
MSPE	<u>7.6072</u>	11.9580	28.0715	1.1405	<u>3.5460</u>	7.4350	9.3596	30.0334	0.2979	3.3619
ROP	6.9671	8.6102	29.7497	0.9303	3.1915	7.3433	8.6900	30.8813	0.3032	3.3219
CLUIE	7.4678	<u>13.6761</u>	28.9923	0.6917	3.4097	7.3512	9.2825	27.6582	0.2975	3.3116
DNIM	**7.6548**	12.4956	28.2592	1.0418	**3.5490**	**7.7030**	12.9647	28.3431	0.3800	<u>3.5347</u>
FUnIE-GAN	7.4456	13.6505	29.7144	0.7349	3.4118	6.8745	5.4877	22.6481	0.4529	2.9861
UWCNN	6.8907	7.5773	27.1204	0.6933	3.1341	6.9417	6.2497	22.1228	0.5012	3.0462
SPV	7.2213	**17.7436**	**33.8266**	0.7008	3.2446	<u>7.6590</u>	**17.2424**	**34.9457**	**0.2218**	**3.5471**

Table 3
Computation time of different methods on Color-Check7, Test-C60, OceanDark, and Test-U45 Datasets.

Methods	Environment	Color-Check7	Test-C60	OceanDark	Test-U45
CBF	Matlab/CPU	14.2777	265.9626	853.4786	13.3836
EUIVF	Matlab/CPU	4.2682	61.4618	267.7903	5.0886
DEA	Matlab/CPU	5.3671	61.6114	226.6983	13.4984
WCD	Matlab/CPU	4.1684	58.5551	218.8312	4.6194
UTV	Matlab/CPU	25.1243	419.1780	1328.9879	17.1879
UDCP	Matlab/CPU	16.7143	252.3891	872.1651	15.8464
UT	Pytorch/CPU	8.3219	77.2854	283.5247	59.6993
TSA	Matlab/CPU	2.3631	25.0400	82.5346	2.1585
MSPE	Matlab/CPU	10.7602	161.6815	584.1737	7.2389
ROP	Matlab/CPU	1.9590	15.7630	62.9100	3.0740
CLUIE	Pytorch/CPU	9.0161	250.5647	669.9133	12.5311
DNIM	Matlab/CPU	10.6005	175.5321	767.4288	14.4073
FUnIE-GAN	Pytorch/CPU	3.0082	69.9870	94.8405	50.0772
UWCNN	Pytorch/CPU	4.1144	126.0898	469.2305	12.8362
SPV	Matlab/CPU	9.6563	92.7824	289.3491	14.5385

techniques in several key areas, including image enhancement, color quality, and artifact minimization. The SPV method excels in five key metrics, securing first place in two categories, with strong performance in the other three. It stands out for its outstanding color quality and minimal artifact presence, while also delivering competitive results in image enhancement, variance preservation, and color fidelity. This indicates that the SPV algorithm has significant advantages in image enhancement capability and stability under low-light and complex underwater environments, effectively improving image quality and detail clarity.

4.5. Qualitative and Quantitative Comparisons on the Test-U45 dataset

1) Qualitative Comparisons

We conducted qualitative comparison experiments on the Test-U45 dataset to evaluate the performance of different algorithms in terms of color correction, contrast enhancement, and visibility improvement in poorly lit and complex underwater environments. As shown in Fig. 8, the CBF (b) method attempts to balance color, but the overall appearance remains somewhat dim, with insufficient detail clarity. EUIVF (c) improves image contrast to some extent, but its color rendering tends to be unnatural. DEA (d) is sensitive to foreground-background depth variations, leaving residual haze in the image. WCD (c) suffers from over-correction in color restoration, frequently introducing unnatural bluish-purple tones. UTV (f) demonstrates unstable performance, resulting in the entire image being dominated by blackness and severe strip-like artifacts. UDCP (g) fails to effectively address red light attenuation, resulting in bluish-green images and possible noise due to excessive contrast enhancement. UT (h) tends to produce halo artifacts along edges during sharpening, making the image appear unnatural. TSA (i) exhibits a noticeable hazy phenomenon, rendering the image with an unnatural cyan-green tone. Moreover, methods such as MSPE (i), ROP (k), and CLUIE (l) still lose some details under extreme lighting conditions, with noticeable color distortion. DNIM (m) performs overall grayish and lacks contrast, with weak color restoration and insufficient detail richness. FunIE-GAN (n) and UWCNN (o) generates a hazy effect and fails to correct the blue-green color shift typically found in underwater images. The SPV (p) method, on the other hand, achieves a more balanced enhancement by preserving natural color while effectively restoring brightness and detail, demonstrating greater stability and realism.

2) Quantitative Comparisons

We systematically evaluated our method on the Test-U45 dataset to comprehensively validate the algorithm's underwater image enhancement performance in addressing color bias, low contrast, and hazy effects. As shown in Table 2, the SPV method stands out with its exceptional performance across key metrics. It achieves the highest UCIQE score of 34.9457, demonstrating superior color quality restoration. Additionally, it excels in image enhancement with an IE of 7.6590 and a low FADE score of 0.2218, indicating minimal artifacts. The method also achieves an impressive VAR of 17.2424, showcasing strong variance preservation. In comparison, other methods exhibit lower performance. SPV consistently delivers high results in image enhancement, color fidelity, variance preservation, and artifact reduction, making it a robust choice for underwater image enhancement.

4.6. Ablation study

We conducted ablation experiments on four datasets: Color-Check7, Test-C60, OceanDark, and Test-U45, by sequentially removing modules from the algorithm. The aim was to verify the effectiveness and contribution of each module in the proposed algorithm from both qualitative and quantitative dimensions. These experiments allowed us to thoroughly evaluate the performance and contribution of each module under different environments and conditions, ensuring the reliability and superiority of the algorithm in practical applications.

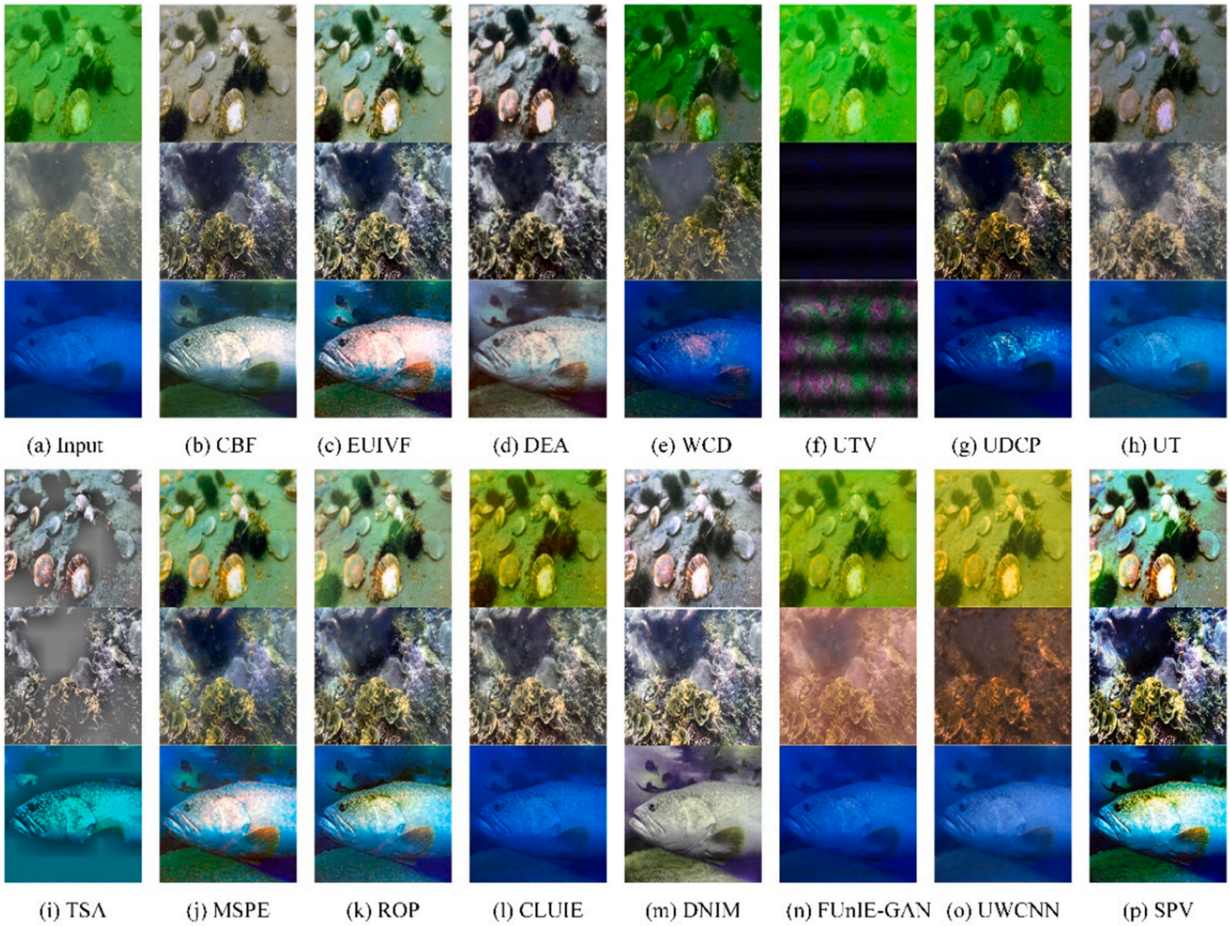

Fig. 8. Comparison of enhancement effects between the SPV algorithm and various underwater image enhancement algorithms on the Test-U45 dataset.

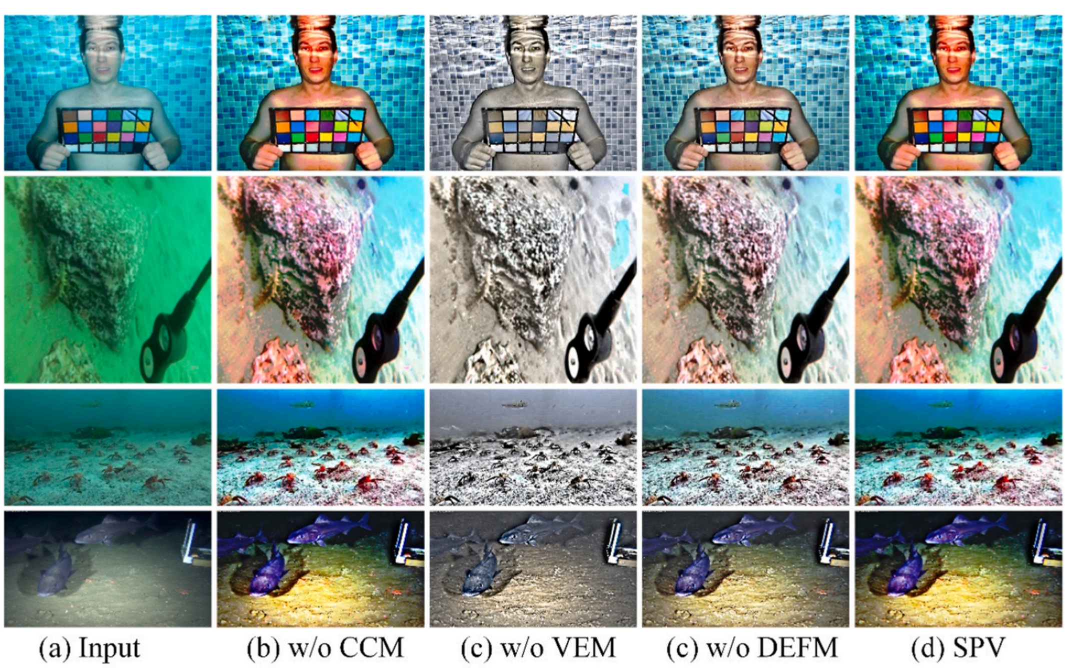

Fig. 9. The enhancement effects of different ablation models in the ablation experiments. We randomly selected images from the Color-Check7, Test-C60, OceanDark, and Test-U45 datasets for evaluation.

In the qualitative ablation experiment evaluation, we randomly selected images from the Color-

Check7, Test-C60, OceanDark, and Test-U45 datasets to assess the performance of these models. The results of the qualitative ablation experiments are shown in Fig. 9. It is evident that the original input images performed poorly under low light and complex environments, exhibiting noticeable color bias and low contrast issues. After removing the color correction module (w/o CCM), although the contrast improved, the overall color deviation was substantial, making it difficult to restore the true scene. After removing the visibility enhancement module (w/o VEM), the images showed poor performance in color restoration and contrast enhancement, with significant detail loss. Finally, after removing the detail enhancement fusion module (w/o DEFM) and replacing it with a 0.5 weighted fusion, the generated enhanced images showed poor color restoration and a loss of contrast compared to SPV. The complete SPV algorithm performed the best in all test images, successfully correcting color bias, enhancing contrast, and excelling in detail preservation. The experimental results demonstrate that the SPV algorithm has significant image enhancement advantages in low-light and complex environments, validating its effectiveness and stability in practical applications.

We also conducted quantitative ablation experiments on the Color-Check7, Test-C60, OceanDark, and Test-U45 datasets, with the results shown in Tables 4 and 5. The experimental results indicate that the performance of the complete model was superior to the ablation models in almost all four datasets. This demonstrates that each key component of the proposed method plays a crucial role in ensuring the overall performance and effectiveness of the algorithm. In summary, the SPV algorithm, through comprehensive ablation experiments, has validated its image enhancement effectiveness in complex underwater environments, showcasing significant advantages and potential for application.

4.7. Application tests

In this section, we further evaluate the performance of the proposed method using three different types of application tests: edge detection (Torre and Poggio, 1986), SIFT feature point matching (Lowe, 2004), and geometric transformation estimation (Lucas and Kanade, 1981). These methods comprehensively assess the robustness, accuracy, and applicability of the method. Through edge detection, we can test the method's stability under different image noise and contrast conditions, evaluating its robustness in edge detection tasks. SIFT feature point matching allows us to test the method's accuracy in image feature point matching, assessing its matching precision under varying scales, rotations, and lighting conditions. Geometric transformation estimation enables us to test the method's applicability in geometric transformation estimation, evaluating its performance in image registration, stitching, and transformation tasks. Additionally, we employed generalizability application tests to further verify the method's generalization ability across different application scenarios. These tests provide a comprehensive understanding of the proposed method's performance in various image processing tasks, thereby validating its effectiveness and reliability in practical applications.

1) Application testing of underwater robots for advanced vision tasks

In this section, we evaluated the proposed method using edge detection, SIFT feature point matching, and geometric transformation estimation. The results for edge detection are shown in Fig. 10. Compared to other algorithms, the images enhanced by the SPV algorithm have clearer edges and effectively suppress noise while preserving edge details, making the edge detection results more robust. The results for SIFT feature point matching are shown in Fig. 11. Compared to other algorithms, SPV ranks among the top in the number of matched feature points, showing a high number of matched feature points. This indicates that the SPV algorithm has higher efficiency and accuracy in feature point detection and matching. The performance of SPV in geometric transformation estimation is also outstanding, as shown in Fig. 12. The results show a high number of matched feature points, indicating that the SPV algorithm ranks among the top in accurately calculating transformation parameters, ensuring precise image registration and transformation.

4.8. Failure case

The proposed method, like other underwater image enhancement techniques, can struggle in certain extreme scenarios, leading to poor contrast, limited visibility recovery, and noticeable color shifts. As illustrated in Fig. 13, neither our algorithm nor current state-of-the-art methods effectively restore these challenging images. The enhanced results still exhibit insufficient visibility, weak contrast, and severe color distortions. This limitation arises because the images contain very little useable information and are captured in highly complex underwater environments characterized by severe light attenuation, dense suspended particles, and significant scattering and absorption effects. These factors pose substantial challenges that current enhancement algorithms cannot fully overcome. Addressing these failure cases requires further quantitative analysis and the development of targeted solution strategies. This remains a key focus for future research, particularly for underwater image enhancement in conditions of low light, extreme attenuation, and high turbidity resulting in very low visibility.

5. Discussion

In this paper, we propose an innovative underwater image enhancement method that combines both physical and non-physical models, aimed at addressing common challenges in underwater image processing, including color bias, low contrast, poor visibility, and lighting compensation. Our method integrates the precision of physical models with the flexibility of non-physical models by using fusion stragety, successfully improving image clarity, contrast, and color accuracy, thereby enhancing visual quality in various underwater environments. Specifically, we introduce a spectrum-based dehazing method that estimates the transmission rate and ambient light in underwater environments to perform initial image dehazing, particularly in complex conditions, restoring image clarity and contrast to ensure high-quality results even in challenging underwater settings. In addition, we utilize a color correction model based on Lab color space, morphological operations, and a visibility restoration module using fuzzy c-means clustering to further enhance color fidelity and visibility. These techniques effectively address color bias, enhance edge details, and significantly improve overall visibility. Finally, our detail enhancement fusion

Table 4
The results of the ablation experiments of the different modules were evaluated on the Color-Check7 and Test-C60 datasets with no-reference evaluation metrics.

Ablated Models	Color-Check7					Test-C60				
	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑
w/o CCM	7.7834	16.0005	34.5629	0.2833	3.6264	6.8305	12.9492	30.5819	0.8695	3.0512
w/o VEM	7.5801	10.5638	28.2543	0.2829	3.4993	6.7581	6.6161	21.0351	1.5834	3.0735
w/o DEFM	7.8554	15.5590	33.5639	0.2206	3.6523	7.3469	14.6174	32.7000	0.6545	3.3651
Full model	7.8613	16.1069	36.2358	0.2339	3.6679	7.2547	15.8880	35.3324	0.6809	3.2595

Table 5
The results of the ablation experiments of the different modules were evaluated on the OceanDark and Test-U45 datasets with no-reference evaluation metrics.

Ablated Models	OceanDark					Test-U45				
	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑	IE↑	VAR↑	UCIQE↑	FADE↓	CEIQ↑
w/o CCM	7.3624	25.1970	32.5444	0.9864	3.3369	7.3542	14.9916	32.6421	0.2768	3.3579
w/o VEM	7.4805	13.4088	27.1832	0.8322	3.4485	7.3003	10.2609	25.2925	0.3722	3.3072
w/o DEFM	7.1927	14.7683	31.7142	0.6170	3.3099	7.6777	17.8606	33.4065	0.1944	3.5823
Full model	7.2213	17.7436	33.8266	0.7008	3.2446	7.6590	17.2424	34.9457	0.2218	3.5471

Fig. 10. Edge detection. The performance of different methods, the proposed method has clearer edges and effectively suppress noise.

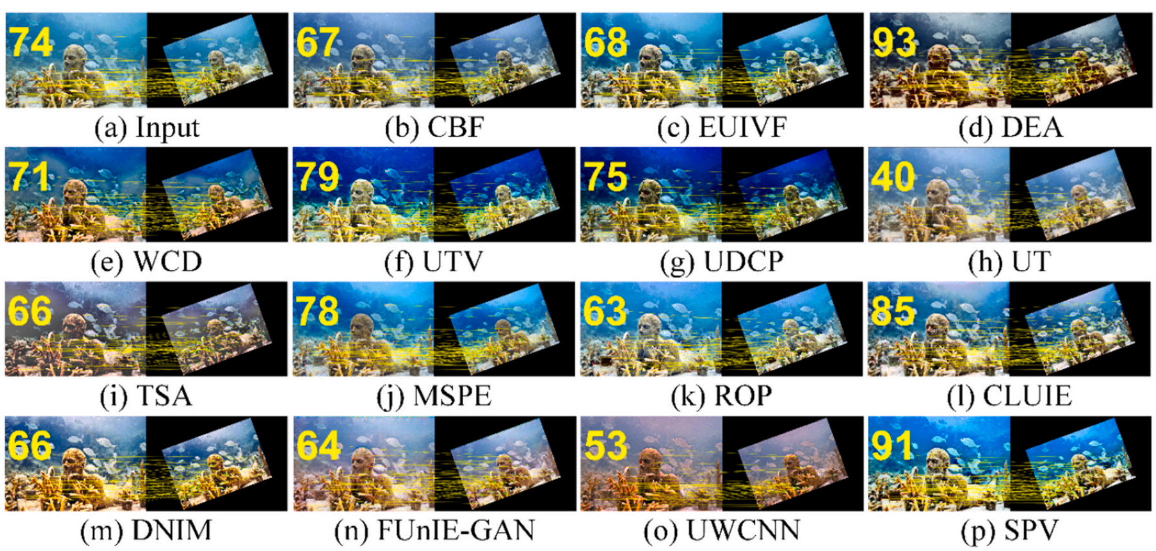

Fig. 11. SIFT feature point matching. The performance of different methods, the proposed method ranks among the top in the number of matched feature points.

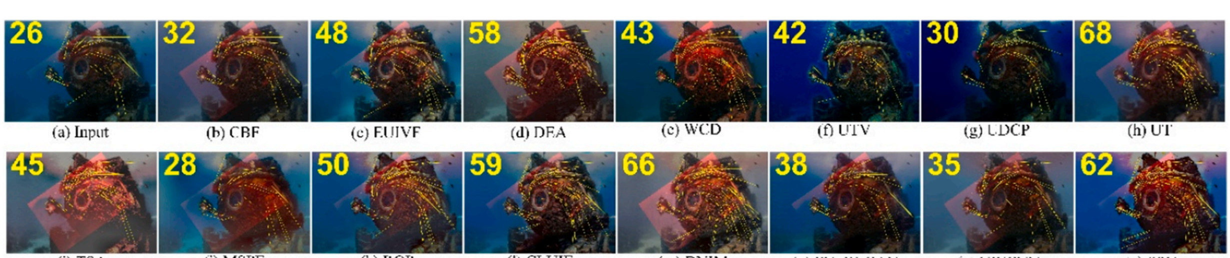

Fig. 12. Geometric transformation estimation. The performance of different methods, the proposed method ranks among the top in accurately calculating transformation parameters.

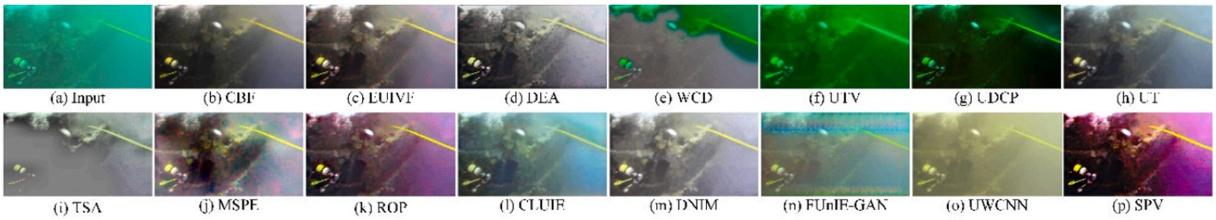

Fig. 13. Failure Case. The performance of different methods under extremely low visibility and poor lighting conditions.

method integrates pixel intensity and global gradients from two different enhancement strategies, achieving complementary advantages and further enhancing image features.

In comparison with 14 advanced underwater image enhancement algorithms across four no-reference image datasets, our method achieves excellent results on multiple key performance metrics. In practical application tests, our method excels in edge detection, SIFT feature point matching, and geometric transformation estimation, demonstrating its robustness and effectiveness in real-world underwater image processing.

Despite these significant achievements, the proposed method still has limitations in certain extreme underwater environments, such as severe light attenuation and high turbidity, where image enhancement results have not reached optimal levels. These conditions require further optimization of the enhancement process, particularly in scenarios with very low visibility. Furthermore, in the future, our method can not only be widely applied in underwater image enhancement but can also be extended to other advanced vision tasks, such as image segmentation (Elmessery et al., 2024; Shams et al., 2025), ecological health assessment (Eliwa and Abd El-Hafeez, 2025) in underwater environments, underwater face recognition (Taha et al., 2023; Ali et al., 2019; Saabia et al., 2019; Eman et al., 2023), underwater robot control (Tamil Thendral et al., 2022; Subhashri et al., 2025) and medical imaging (Eliwa et al., 2024; Mostafa et al., 2024; Abd El-Hafeez, 2010; Omar and Abd El-Hafeez, 2024). This will further enhance its effectiveness and applicability across a range of practical applications.

6. Conclusion

The underwater image enhancement method proposed in this paper has achieved significant improvements in several key metrics, especially demonstrating outstanding performance on the Test-U45 and Color-Check7 datasets. Compared to the second-best methods, our approach improved UCIQE by 10.63 %, and VAR by 16.10 % on the Test-U45 dataset, and improved UCIQE by 10.15 %, and VAR by 12.50 % on the Color-Check7 dataset. These results indicate that our method significantly outperforms existing advanced underwater enhancement algorithms in terms of image clarity, contrast, and color restoration, especially showing notable advantages in UCIQE, CEIQ, and VAR metrics. Moreover, SPV also achieved good results on other datasets. Despite facing challenges in certain extreme conditions, it has demonstrated excellent robustness and effectiveness, especially in real-world applications such as edge detection, feature matching, and geometric transformation estimation.

In the future, we will further optimize the image enhancement process, particularly in conditions of high turbidity and low visibility. Additionally, our method is not only applicable to underwater image enhancement but also extendable to other advanced visual tasks such as object detection, image segmentation, and augmented reality, further improving its performance and applicability in various real-world applications.

CRediT authorship contribution statement

Qifeng Liu: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Xin Yan:** Validation, Resources, Formal analysis. **Lu Shen:** Investigation, Data curation. **Qiang Li:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledge

The authors would like to express their deepest gratitude to Codruta O. Ancuti, Chongyi Li, Tunai Porto Marques, and Hanyu Li for providing the standard datasets of real underwater images, which were essential for the evaluation and validation of the algorithm. The authors also wish to extend their sincere thanks to the students and teachers who assisted in this work, offering valuable suggestions and guidance.

Data availability

Data will be made available on request.

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